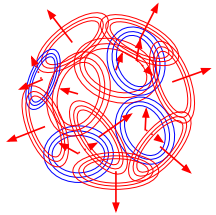
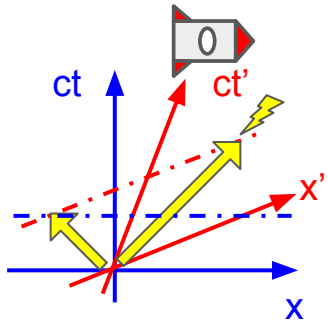
R-space Theory 1.0

- Geometric Integration of Relativity and Quantum Mechanics -



Chap.1:
Why is the Speed of Light Always Constant?



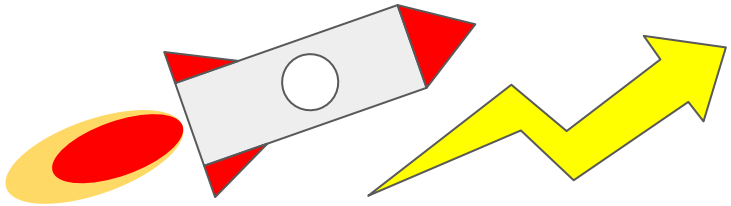
1. Why is the Speed of Light Always Constant?

The speed of light remains constant even if there is relative velocity between the light source and the observer.

Furthermore, no object can exceed the speed of light.

Why is the speed of light always constant?

The answer is simple: it is an observational fact in our world.

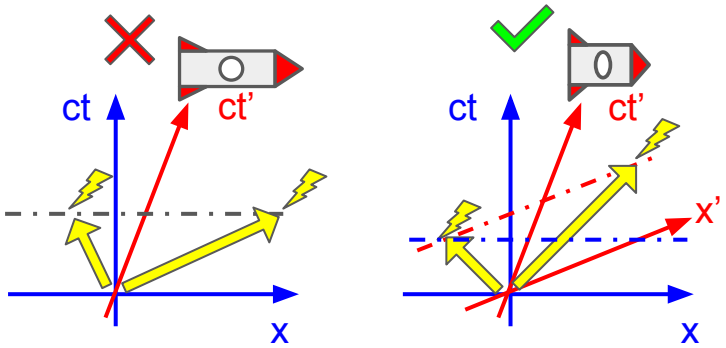


1. Why is the Speed of Light Always Constant?

When moving at speeds close to that of light, time dilation and length contraction occur, phenomena confirmed through various experiments and explained by Einstein's principle of the constancy of the speed of light.

However, Einstein's principle is a "principle" and does not explain why the speed of light is constant.

Let us consider the underlying reason.

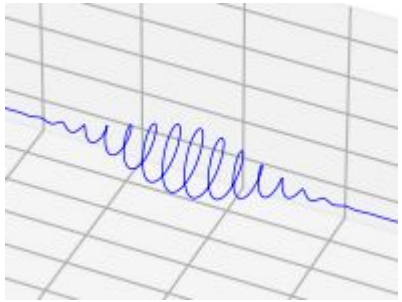


2. The World is Like a Collection of Vortices

While no object can exceed the speed of light in our macroscopic world, the quantum world is different.

Quanta exist probabilistically and can, in a probabilistic sense, exist at an infinite distance in an instant.

To resolve this contradiction, R-space Theory re-examines the nature of spacetime.

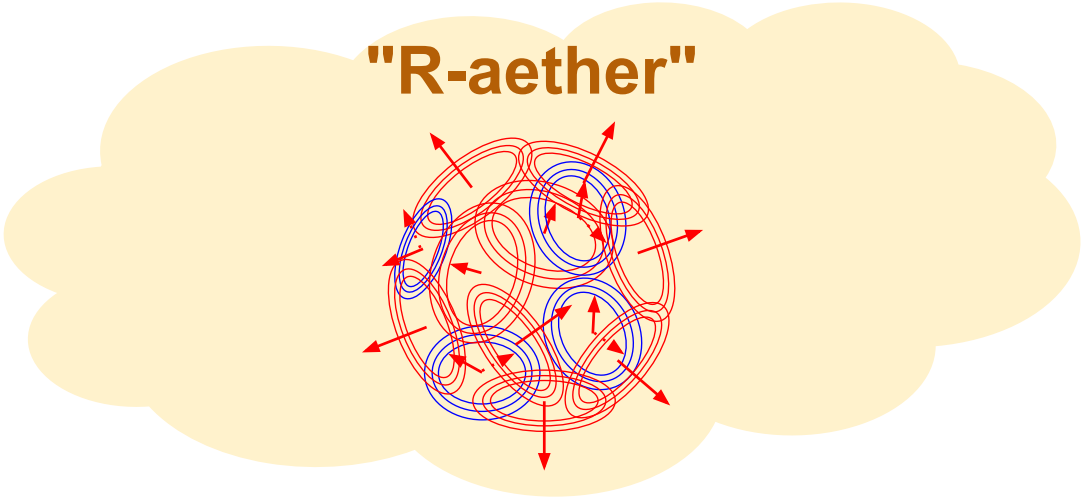


2. The World is Like a Collection of Vortices

In R-space, the time and space we recognize do not exist. Instead, the world is composed of vortices generated in a medium called "R-aether."

We observe these vortices as the time and space of our perception.

Next, I will explain how our world is constructed from these vortices.



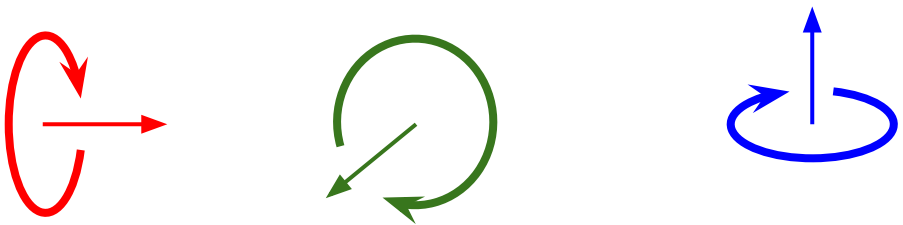
3. Rotation and Phase

In R-space, there is no distance or time; there is only rotation and phase.

The direction of rotation has three degrees of freedom, corresponding to the three dimensions (x, y, z) of our world, represented by red, green, and blue.

The arrows at the center indicate the direction and magnitude of the rotation, which we call the rotation vector.

This rotation vector possesses rotational energy.



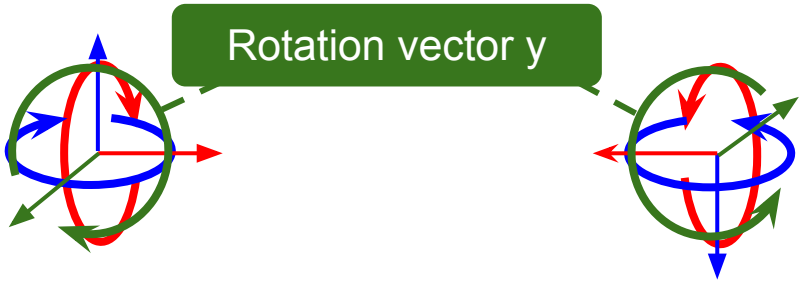
3. Rotation and Phase

A photon, a particle of light, is a vortex with three rotational components: x, y, and z.

Consider two photons colliding in R-space.

Since R-space only consists of rotation and phase, the collision is between phases.

For example, consider two photons colliding at their y-components of the rotation vector.

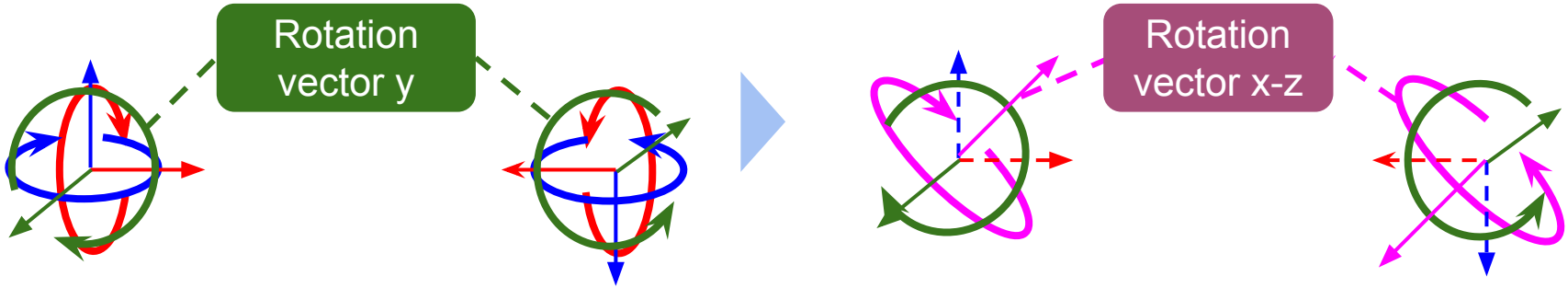


3. Rotation and Phase

Viewed from rotation vector y, the x and z vectors are orthogonal.

Their composite, the rotation vector x-z (shown in magenta), is also orthogonal to the y vector.

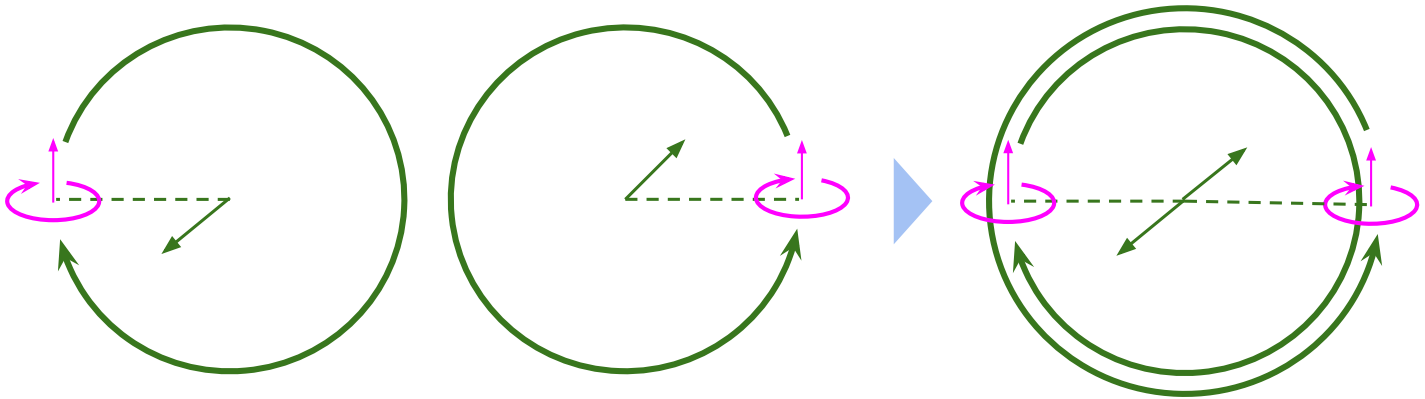
The photon can thus be simplified into two rotations: vector y and vector x-z.



3. Rotation and Phase

To make the phase progress clear, we can represent rotation vector x-z as a phase positioned on the circumference of a larger circle representing rotation vector y.

The two photons are then superimposed.

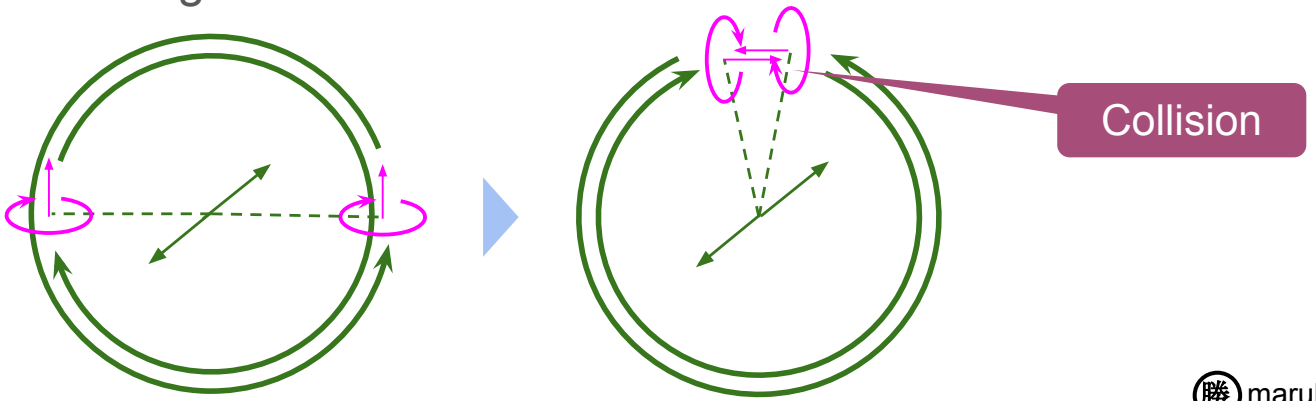


3. Rotation and Phase

When the phase progress of the y-vectors of two photons matches, they interact.

If the x-z vectors rotate in the same direction, they pass each other; if opposite, they collide and reverse direction.

Since the two photons are indistinguishable, both cases appear as if the phases passed through each other.

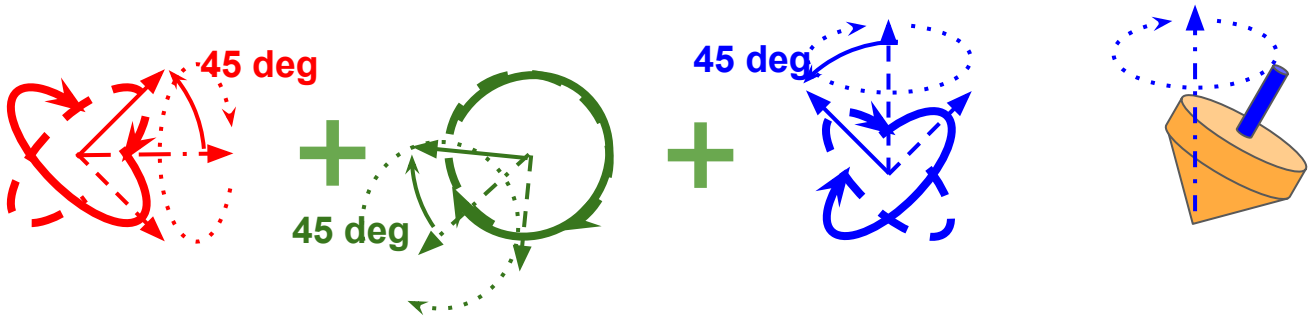


3. Rotation and Phase

This is the world of photons.

However, our bodies and instruments are made of atoms, which consist of electrons and quarks.

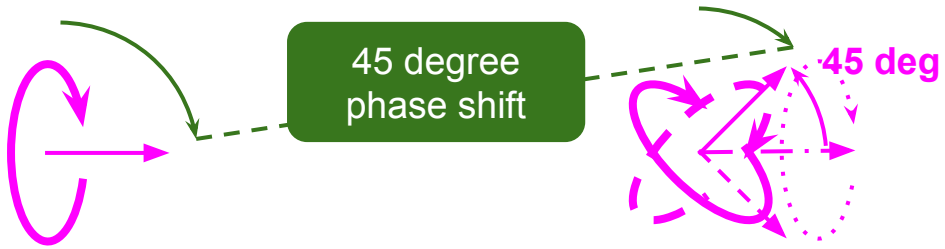
These are also vortices in the R-aether, but unlike photons, their x, y, and z rotation vectors precess like a spinning top due to mutual influence.



3. Rotation and Phase

The angle of precession for electrons and quarks is always 45 degrees.

Consequently, there is a 45-degree phase shift between the rotation axes of photons and those of electrons or quarks.



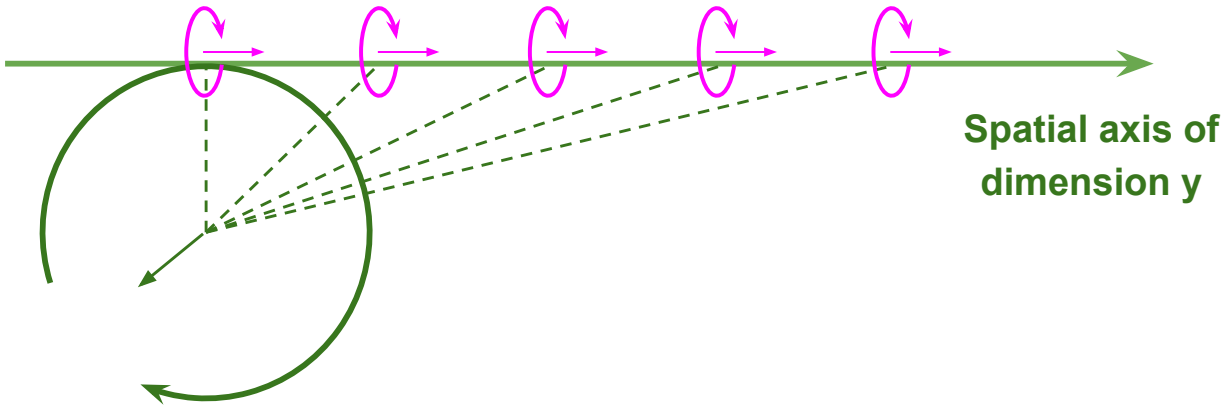
4. How Spacetime is Generated from Rotation Vectors

Draw a tangent line to the circle of rotation vector y.

Then, draw a line from the center of the circle toward the direction of the x-z phase until it hits the tangent line.

This mathematical process is called projection.

This tangent line corresponds to the spatial axis in our spacetime.



4. How Spacetime is Generated from Rotation Vectors

The circumference of rotation vector y is projected across the entire universe's y-axis.

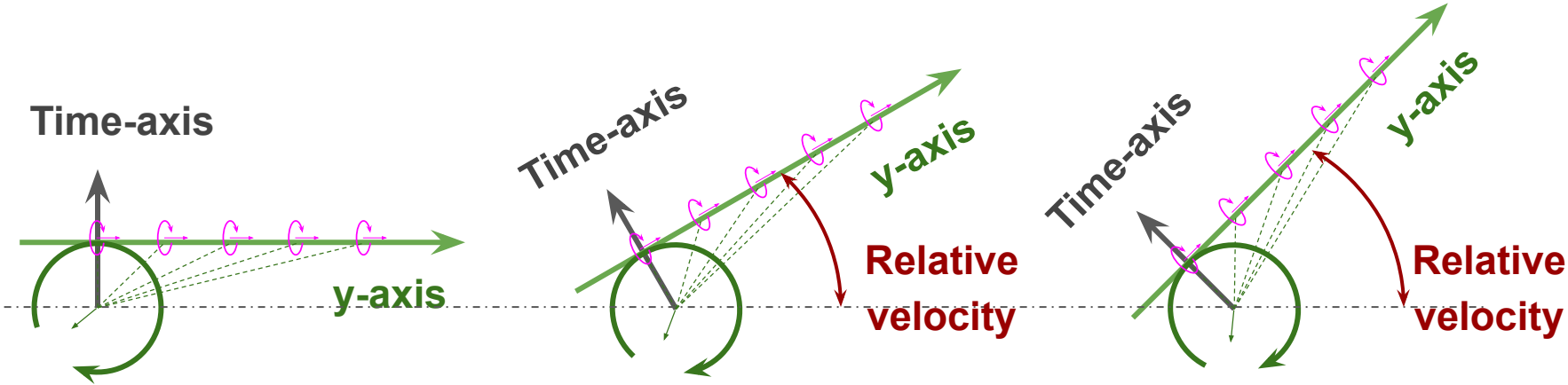
This means the circumference corresponds to the spatial y-axis of the whole universe.

This is similar to how a spherical mirror reflects an entire landscape (but in reverse) or how the Mercator projection expands Antarctica.

In R-space, a photon travels the entire universe every time its phase completes one rotation.

4. How Spacetime is Generated from Rotation Vectors

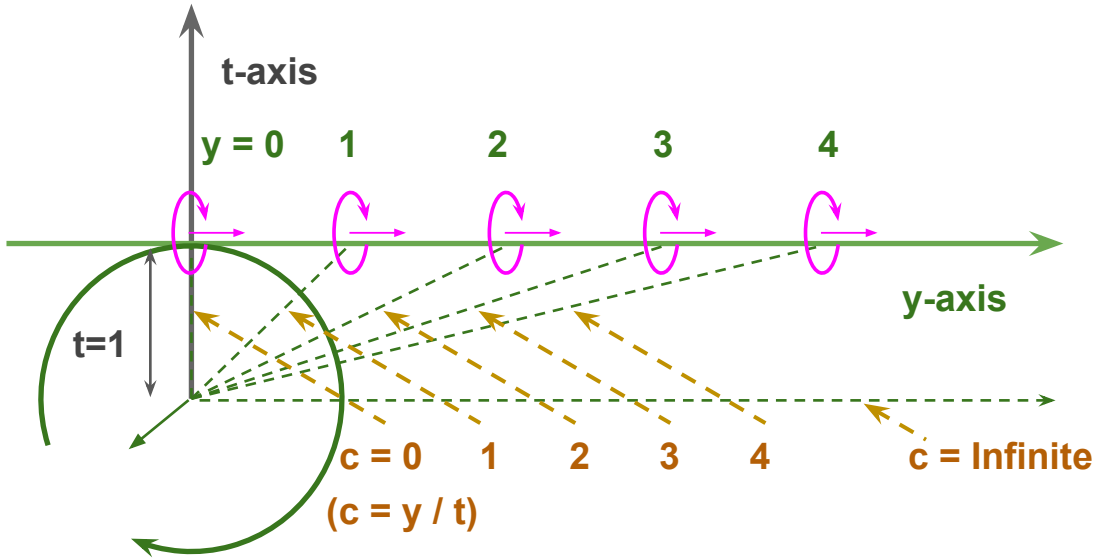
The direction of the tangent line (the spatial axis) depends on the observer's relative velocity.



4. How Spacetime is Generated from Rotation Vectors

The slope of the phase relative to the spatial axis and its orthogonal time axis corresponds to velocity (c=y/t).

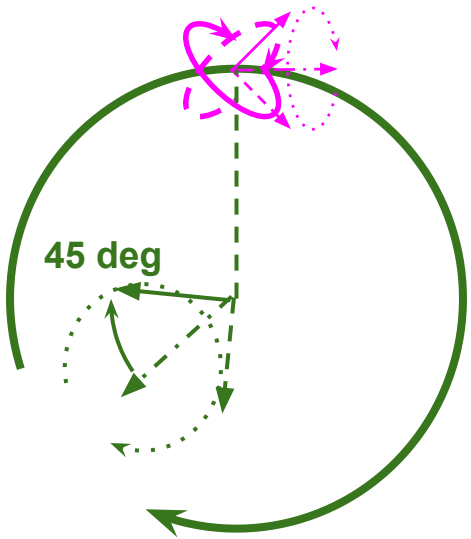
In R-space, light's velocity can range from zero to infinity depending on the phase slope.

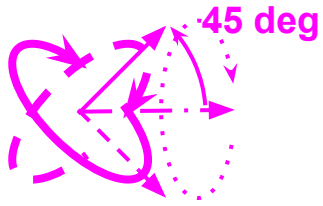


4. How Spacetime is Generated from Rotation Vectors

Why, then, do we observe light and matter moving at a specific constant speed?

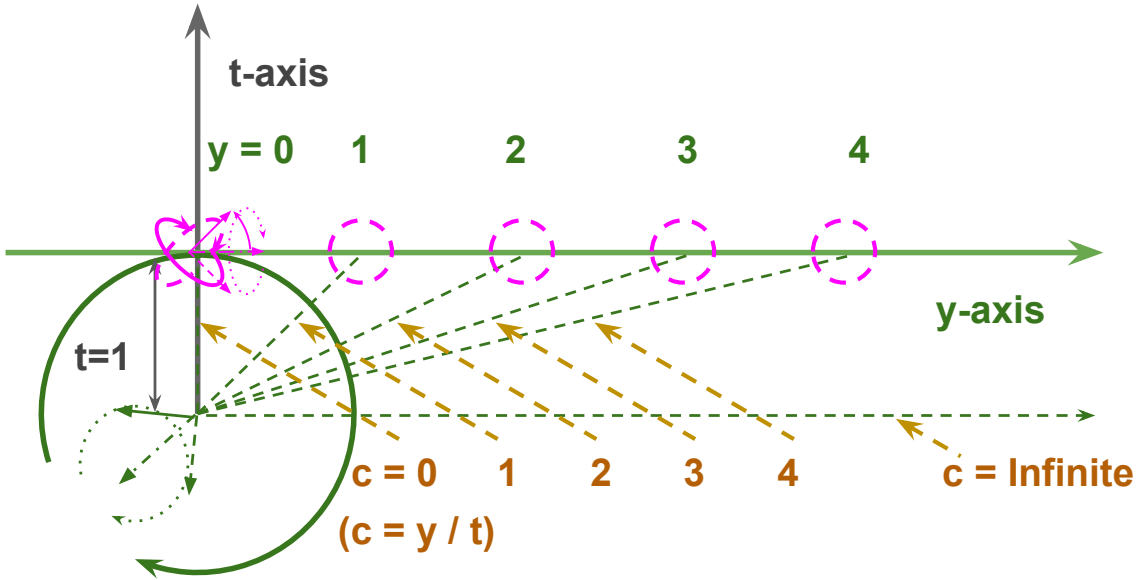
It is because the rotation vectors of electrons and quarks precess at 45 degrees.





5. The Mechanism of the Constancy of the Speed of Light

Like photons, the phases of electrons and quarks can be projected onto a tangent line, theoretically ranging from zero to infinite speed.



5. The Mechanism of the Constancy of the Speed of Light

However, there are differences between photons and electrons and quarks.

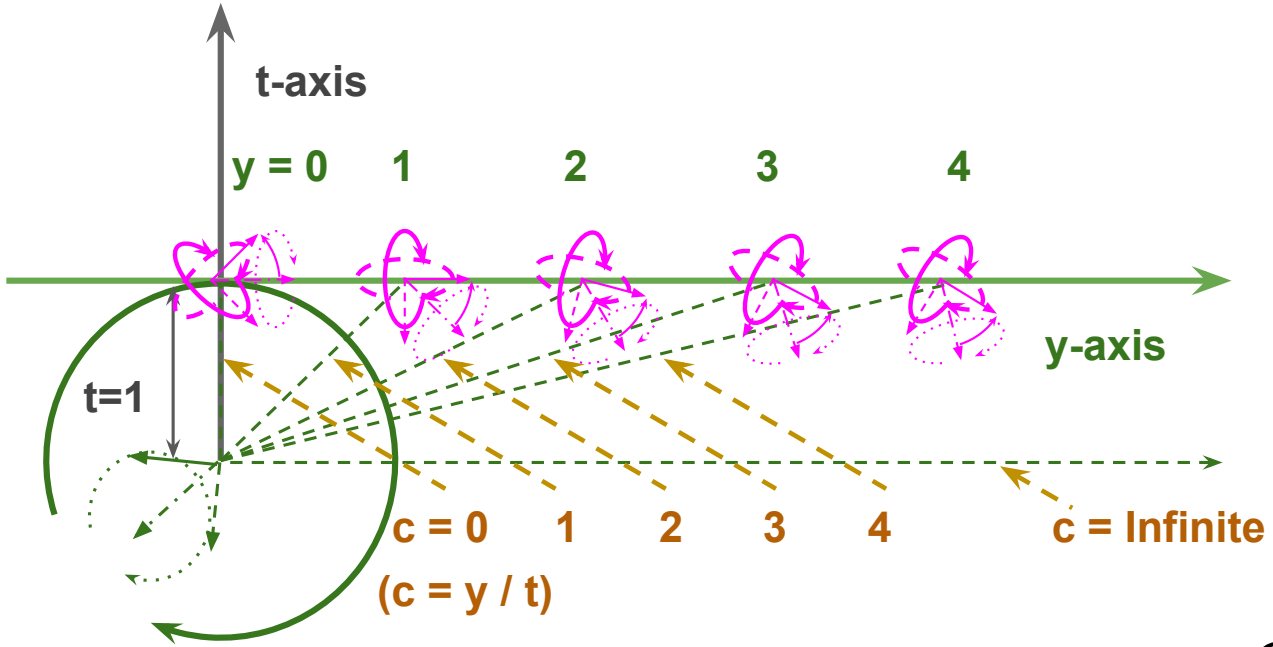
The rotation vector x-z of a photon always points in the y-axis direction.

However, for electrons and quarks, due to the influence of the precession of the rotation vector y-axis, the central axis of the precession of the rotation vector x-z does not always coincide with the y-axis direction.

The central axis of the precession is projected by rotating around a direction perpendicular to the y-axis (the center of rotation of the precession of the rotation vector y).

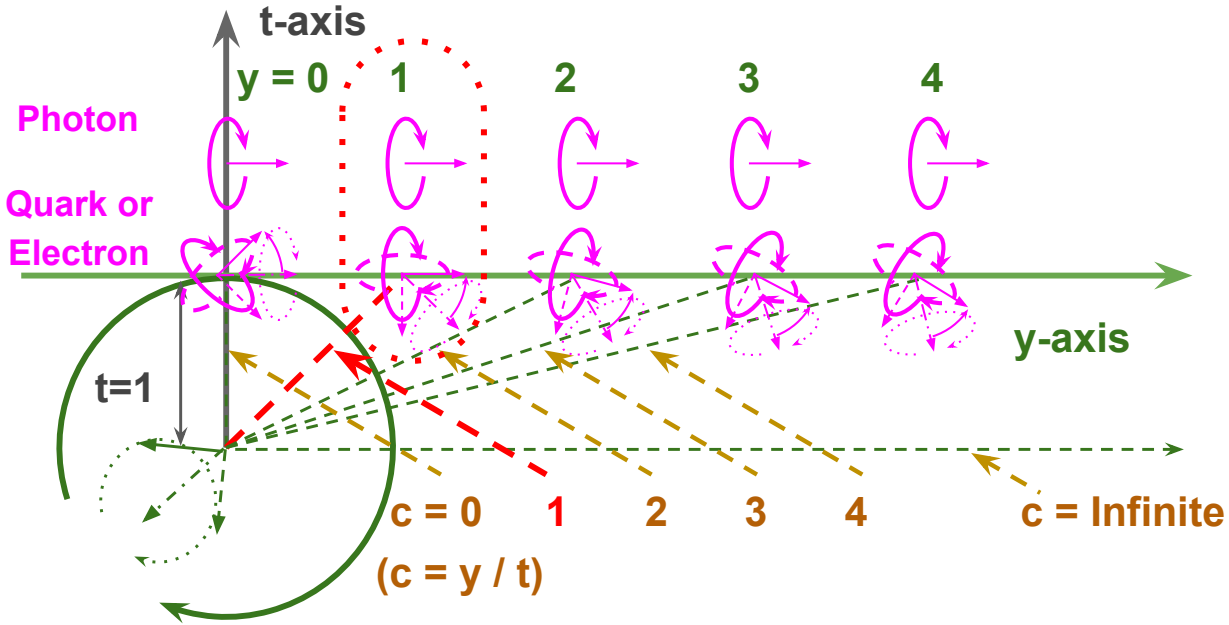
5. The Mechanism of the Constancy of the Speed of Light

Furthermore, the central axis of the precession of the rotation vector x-z rotates while remaining perpendicular to the phase line of the rotation vector y.



5. The Mechanism of the Constancy of the Speed of Light

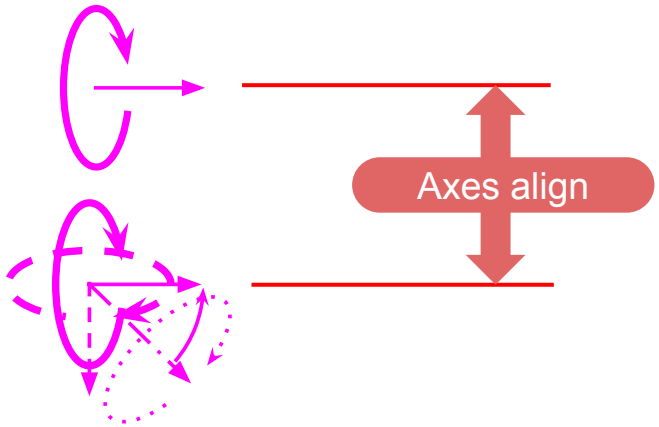
When overlapping the projected states, the rotation axes of the photon and the quark/electron align only when the speed c=1 (at a 45-degree phase difference).



5. The Mechanism of the Constancy of the Speed of Light

Interaction is only possible when these axes align.

This is why, although they possess speeds from zero to infinity in R-space, the speed of light c is always observed as constant.

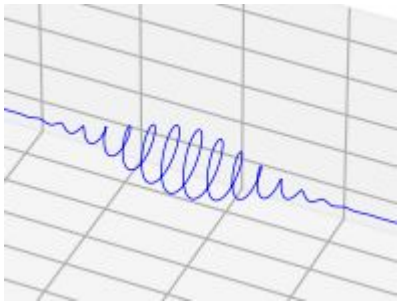


5. The Mechanism of the Constancy of the Speed of Light

Regarding the speed of matter: the precession of electron/quark rotation vectors creates "unevenness" (density variations) in how their phases spread through space.

The movement speed of this unevenness is what we observe as the velocity of objects, which is always slower than light.

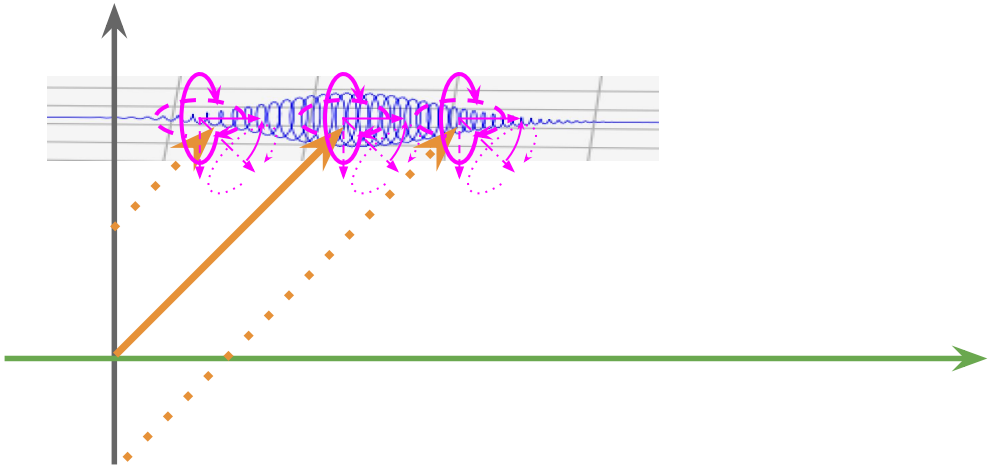
This is why electrons and quarks have mass.



5. The Mechanism of the Constancy of the Speed of Light

Even if phases collide at vast distances due to quantum probability, they always collide at a 45-degree phase difference, ensuring the observed speed of light remains constant.

This resolves the contradiction between the principle of invariant light speed and probabilistic quantum observations.

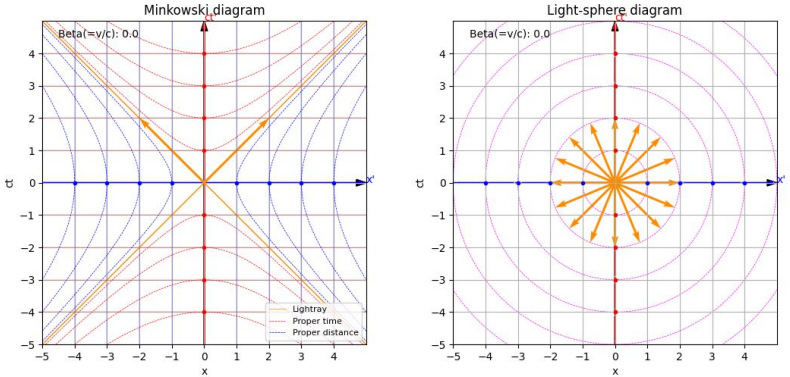


6. Why Does Our Spacetime Look Like Minkowski Space?

R-space and Minkowski space are simply different expressions of the same thing.

While R-space has no "spatial extension" as we know it, the number of rotations is expressed as concentric circles.

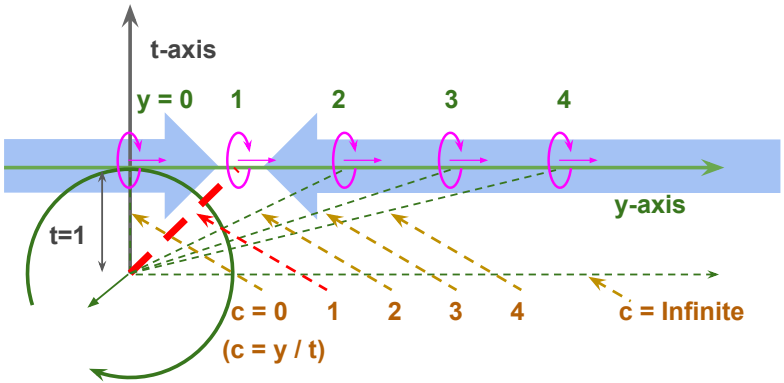
In R-space, light spreads instantly to the whole universe, but it only interacts with matter at a constant speed (45-degree phase difference).

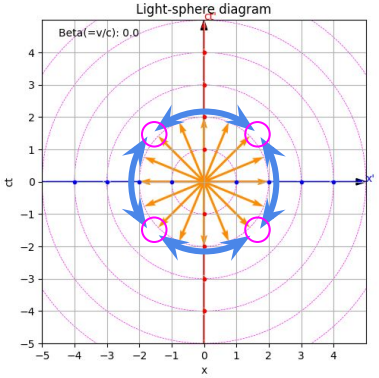


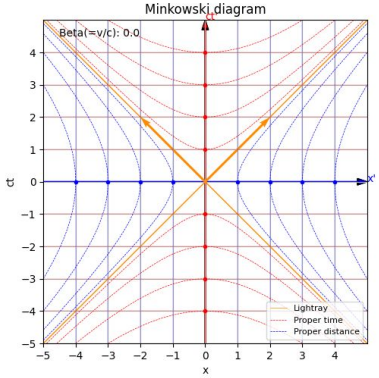
6. Why Does Our Spacetime Look Like Minkowski Space?

Minkowski space converges these infinite speeds into a single velocity (the 45-degree diagonal).

It's like the reverse of a Mercator projection, where an infinitesimal point (the pole) is stretched into a line.



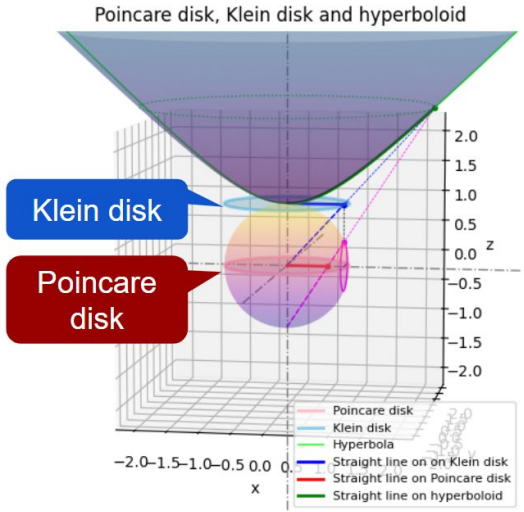




6. Why Does Our Spacetime Look Like Minkowski Space?

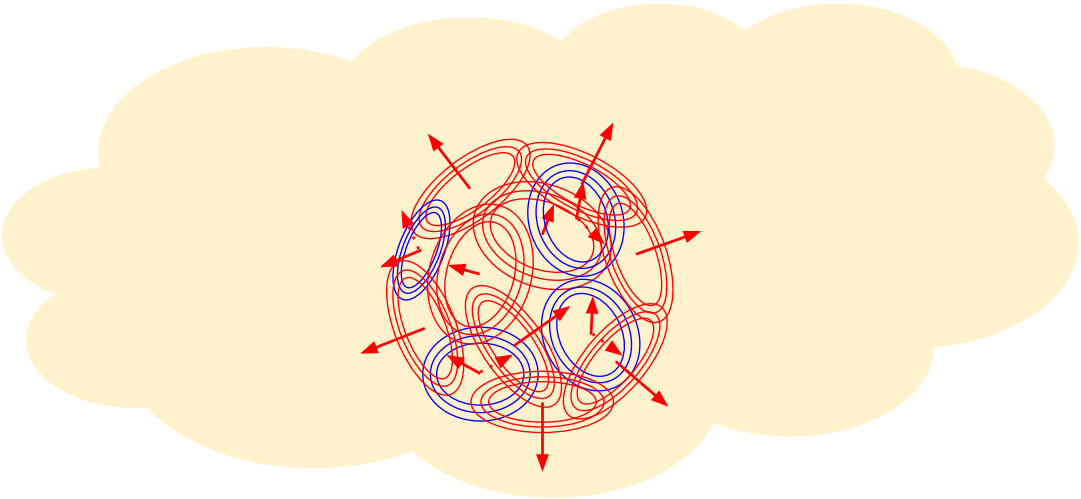
The circles of rotation vector y in R-space are stretched into hyperbolas in Minkowski space.

This relationship is equivalent to a circle being projected into a hyperbola via a Poincaré disk or Klein disk.



6. Why Does Our Spacetime Look Like Minkowski Space?

The universe is made of the rotation of vortices, and their precession is the reason the speed of light is always observed as constant.



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This is all for my presentation.

Thanks for watching!

